Brown Invitational - Repaired practice edition - Answer key

Division B | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

s2-10. (2 points) Authored replacement for source item s2-10: Supplied field notes place celestite with gypsum in evaporite beds. Describe an environment consistent with those observations.

answer: An evaporative basin where dissolved sulfate minerals precipitate as water becomes concentrated; an arid restricted marine basin is one suitable example. Use the supplied field notes, not the unrelated fireworks answer. (2 points).

s2-11. (1 points) Authored replacement for source item s2-11: Supplied crystallographic data: chalcopyrite is tetragonal and celestite is orthorhombic. Do these two specimens belong to the same crystal system?

answer: No. Tetragonal and orthorhombic are different crystal systems. (1 points).

s6-3. (2 points) Authored replacement for source item s6-3: Repair data: green amazonite is a variety of microcline; microcline is triclinic and orthoclase is monoclinic; both feldspars have composition KAlSi3O8. Explain which pair illustrates polymorphism.

answer: Microcline and orthoclase have the same composition and different crystal structures, so they illustrate polymorphism. Amazonite is a variety of microcline, not a separate polymorph contrasted with microcline. (2 points).

s6-5. (2 points) Authored replacement for source item s6-5: Can a green feldspar’s color or ultraviolet appearance alone establish that it contains lead? Describe a stronger test.

answer: No. Color/fluorescence is not a uniquely diagnostic elemental measurement. Use a suitable calibrated chemical/elemental analysis and standards rather than inferring lead solely from an appearance. (2 points).

s15-1. (4 points) Authored replacement for source item s15-1: Label these supplied descriptions: A is a sheet parallel to bedding; B cuts across bedding; C is a concordant dome-shaped intrusion; D is an intrusive body with exposed area greater than 100 square kilometres.

answer: A: sill. B: dike/dyke. C: laccolith. D: batholith. Award one point for each correct label. (4 points).

s16-1. (2 points) Authored replacement for source item s16-1: Name and explain these four relative-dating principles: superposition, original horizontality, lateral continuity and cross-cutting relationships.

answer: Superposition: younger beds overlie older beds in an undeformed sequence. Original horizontality: sediments are initially deposited approximately horizontally. Lateral continuity: beds originally extend laterally until thinning or reaching a boundary. Cross-cutting: the cutting feature is younger than what it cuts. Award credit for the meaning of each specified principle. (2 points).

s17-1. (3 points) Authored replacement for source item s17-1: Replace the missing facies diagram using this supplied schematic: A has moderate pressure and lower temperature; B has similar pressure and higher temperature; C has high pressure and low temperature. The supplied candidate facies are greenschist, amphibolite and blueschist. Match each once.

answer: A: greenschist. B: amphibolite. C: blueschist. These are qualitative positions in a supplied schematic, not exact pressure-temperature boundaries. (3 points).

s17-2. (3 points) Authored replacement for source item s17-2: Explain why hornfels, greenschist, blueschist, amphibolite and granulite cannot all be put into one universal lowest-to-highest temperature grade sequence without pressure information.

answer: Facies depend on pressure as well as temperature. Blueschist is characteristically high-pressure/low-temperature, while hornfels is associated with low-pressure contact metamorphism. At comparable moderate pressure, greenschist-amphibolite-granulite can express increasing temperature, but it is not a universal ordering of all listed facies. (3 points).